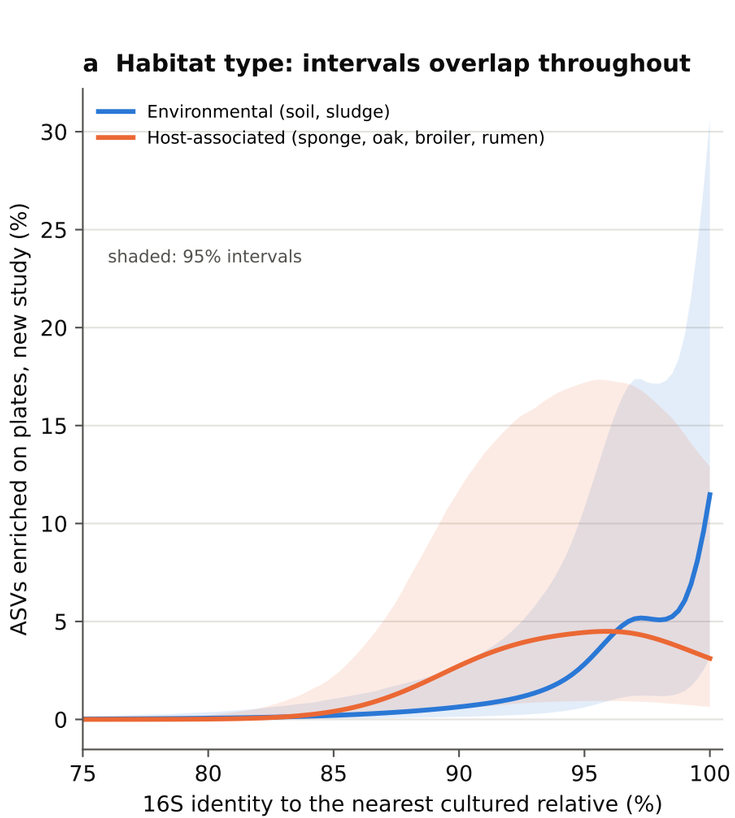
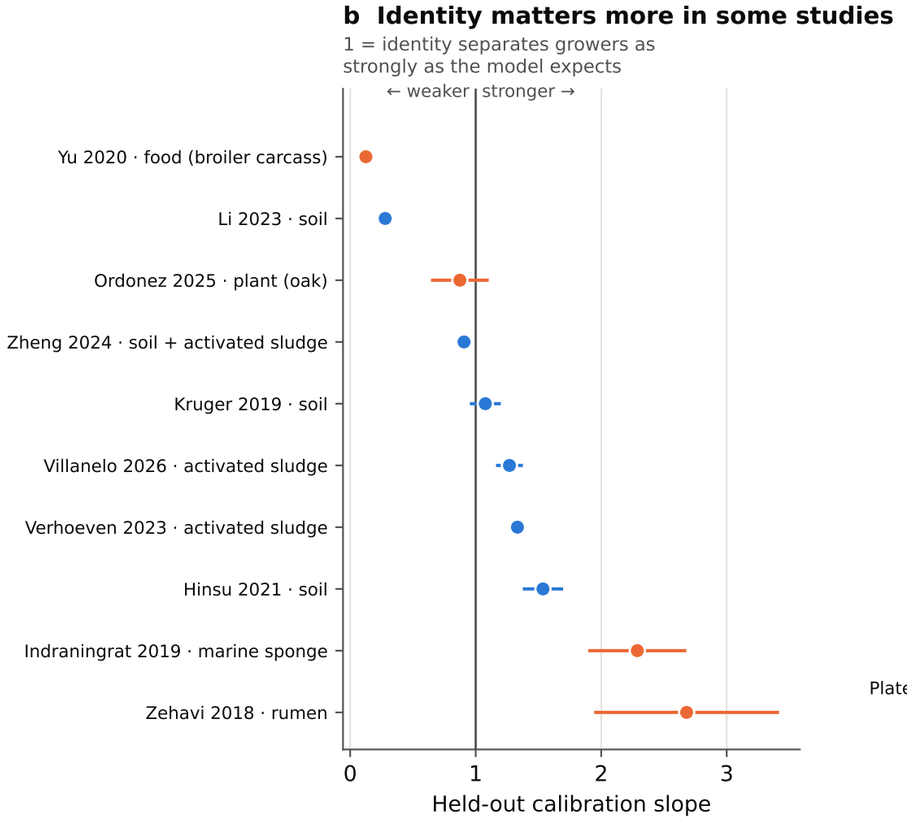
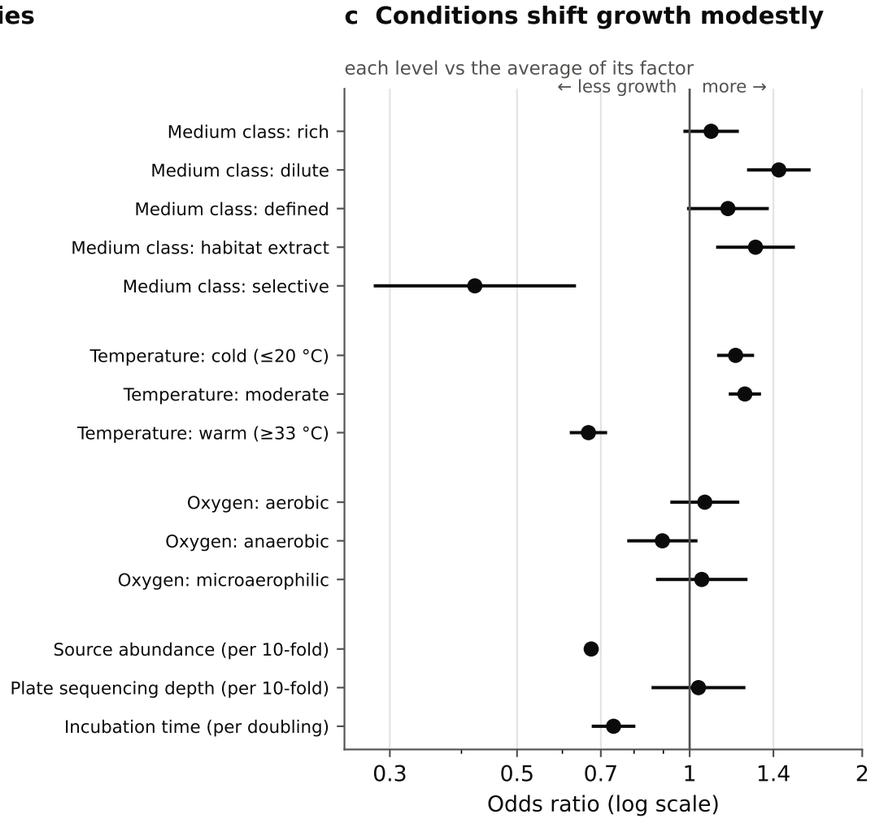

a Habitat type: intervals overlap throughout
Environmental (soil, sludge)
Host-associated (sponge, oak, broiler, rumen)
shaded: 95% intervals
ASVs enriched on plates, new study (%)
16S identity to the nearest cultured relative (%)


b Identity matters more in some studies
1 = identity separates growers as strongly as the model expects
← weaker stronger →
Yu 2020 · food (broiler carcass)
Li 2023 · soil
Ordonez 2025 · plant (oak)
Zheng 2024 · soil + activated sludge
Kruger 2019 · soil
Villanelo 2026 · activated sludge
Verhoeven 2023 · activated sludge
Hinsu 2021 · soil
Indraningrat 2019 · marine sponge
Zehavi 2018 · rumen
Held-out calibration slope


c Conditions shift growth modestly
each level vs the average of its factor
← less growth more →
Medium class: rich
Medium class: dilute
Medium class: defined
Medium class: habitat extract
Medium class: selective
Temperature: cold (≤20 °C)
Temperature: moderate
Temperature: warm (≥33 °C)
Oxygen: aerobic
Oxygen: anaerobic
Oxygen: microaerophilic
Source abundance (per 10-fold)
Plate sequencing depth (per 10-fold)
Incubation time (per doubling)
Odds ratio (log scale)